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ETE 4201

Sensor Project: Temperature Sensor

This project is to be able to control a 4pin PWM fan based on the current ambient temperature of the surrounding area. The project will be able to display the current temperature as well as the speed (RPM) at which the fan is spinning. The controlling aspect is achievable through a temperature sensor's input into a microcontroller to then be able to control the fan based on the code implemented.

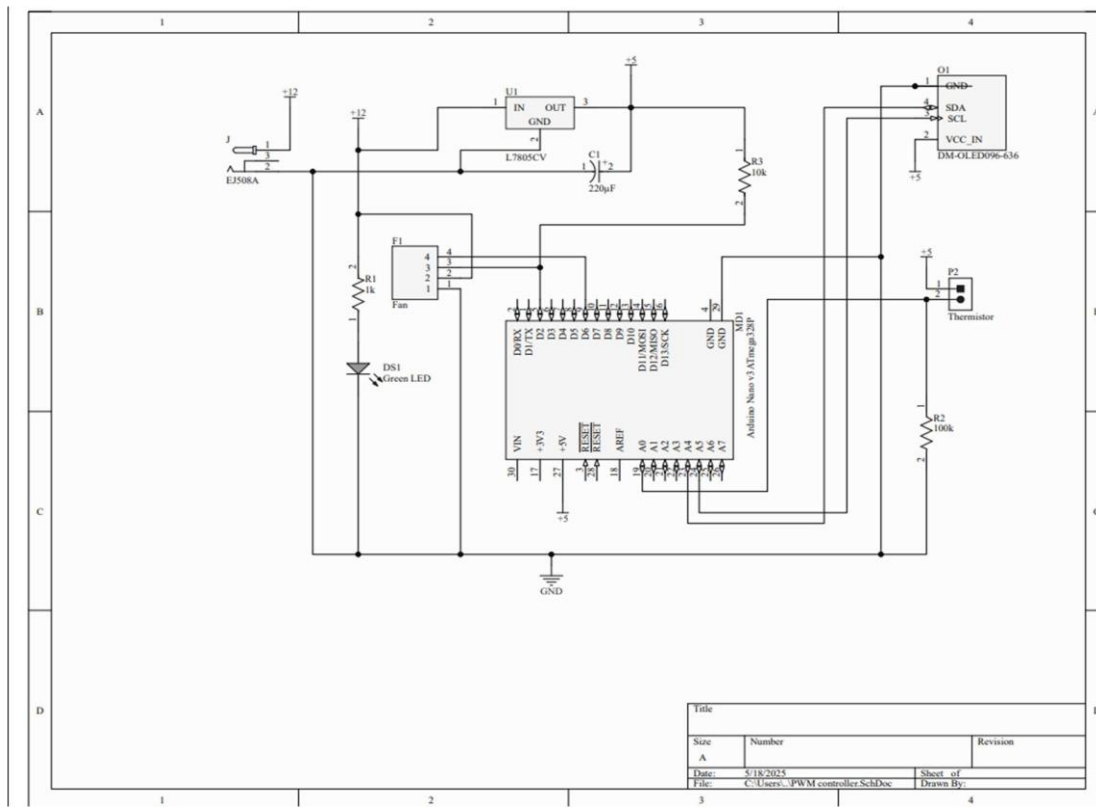
Circuit Operation:

The DC12V barrel jack will supply power the entire component, most specifically for the PWM fan. There will be an LED to show that the PCB is on. The L7805 voltage regulator will step down the voltage from 12V to 5V as the rest of the components only can accept up to 5V. These components are the Arduino Nano V3 and the OLED 0.96" display. The Arduino Nano V3 will be the brains of this project to control the speed of the PWM fan and to display the readings onto the OLED display. The temperature sensor is a 10kOhm Thermistor that is connected to the input of the Arduino nano on pin A0. There are three total resistors used in this project, a 1k ohm resistor and two 10k ohm resistors.

What I learned:

From doing this project, we were able to learn how thermistors work and how to convert their analog readings to actual temperature readings. Example of this is the connections between the connection A0 on the Arduino to the thermistor. By reading the change in resistance, we can calculate the ambient temperature. This allows us to gain experience in Sensor interfacing using a microcontroller, analog-to-digital conversion (ADC), and understanding the temperature relationship of resistance-temperature.

Schematic:



Picture of Temperature Sensor

